CS3491 ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

COURSE OBJECTIVES:

The main objectives of this course are to:

- Study about uninformed and Heuristic search techniques.
- Learn techniques for reasoning under uncertainty
- Introduce Machine Learning and supervised learning algorithms
- Study about ensembling and unsupervised learning algorithms
- Learn the basics of deep learning using neural networks

UNIT I PROBLEM SOLVING

Introduction to AI - AI Applications - agents — uninformed search strategies — Heuristic search strategies — Local search and optimization problems — adversarial search — constraint satisfaction problems (CSP)

UNIT II PROBABILISTIC REASONING

Acting under uncertainty — Bayesian inference — naive bayes models. Probabilistic reasoning — Bayesian networks — exact inference in BN — approximate inference in BN — causal networks.

UNIT III SUPERVISED LEARNING

Introduction to machine learning — Linear Regression Models: Least squares, single & multiple variables, Bayesian linear regression, gradient descent, Linear Classification Models: — Discriminant function — Probabilistic discriminative model - Logistic regression, Probabilistic generative model — Naive Bayes, Maximum margin classifier - Support vector machine, Decision Tree, Random forests

UNIT IV ENSEMBLE TECHNIQUES AND UNSUPERVISED LEARNING

Combining multiple learners: Model combination schemes, Voting, Ensemble Learning - bagging, boosting, stacking, Unsupervised learning: K-means, Instance Based Learning: KNN Gaussian mixture models and Expectation maximization

UNIT V NEURAL NETWORKS

Perceptron - Multilayer perceptron, activation functions, network training — gradient descent optimization — stochastic gradient descent, error backpropagation, from shallow networks to deep networks — Unit saturation (aka the vanishing gradient problem) — ReLU, hyperparameter tuning, batch normalization, regularization, dropout